

Introduction to Bison and Lex

Building a C to XML Parser

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Course Duration: 60 Minutes.

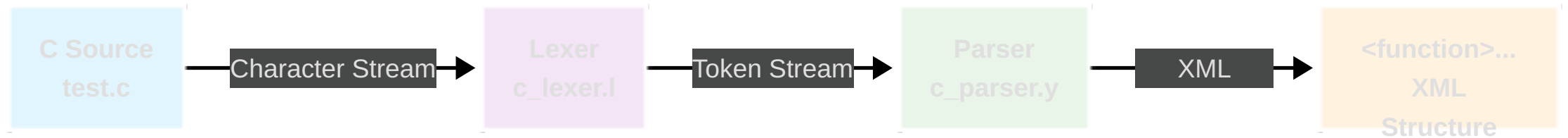
What You Will Learn Today

By the end of this 60-minute session, you will:

- Understand the roles of **Lex** (Flex) and **Yacc** (Bison).
- Know the basic structure of `.l` (Flex) and `.y` (Bison) files.
- Be able to build a simple **lexer** and **parser**.
- Have a foundational codebase to parse a subset of C and output **XML**.

The Big Picture: What Are We Building?

Goal: A program that reads C code and outputs its structure as XML.



Meet The Tools: Lex (or Flex)

- **Lex:** *A lexical analyzer generator.*
- Its job is to break input text into a stream of **tokens**.
- You define tokens using **Regular Expressions**.
- Example Tokens for C: `int`, `while`, `(`, `)`, `{`, `}`, `identifier`, `number`.

Input:

```
int main() {
```

Output (Token Stream):

```
INT IDENTIFIER(main) LPAREN RPAREN LBRACE
```

Meet The Tools: Bison

- **Bison:** *A parser generator.*
- Its job is to find structure in the stream of tokens from Flex.
- You define structure using a **Context-Free Grammar**.
- It checks if the token stream follows the grammar rules of the language (e.g., C).

Grammar Rule:

```
function : INT IDENTIFIER '(' ')' compound_statement ;
```

How They Work Together

1. **Bison** calls `yylex()` to get the next token.
2. **Lex** runs, matches input against its rules, and returns a token to Bison.
3. **Bison** uses these tokens to build the parse tree according to its grammar rules.
4. We insert **C code actions** in Bison to generate XML output when rules are matched.

Course Agenda

1. **Introduction & Overview** (*Now!*)
2. **Hands-On with Lex:** Building the Tokenizer.
3. **Hands-On with Bison:** Defining the Grammar.
4. **Linking Lex and Bison:** Compiling the Parser.
5. **Applied Project:** Creating our C $\rightarrow$ XML parser.
6. **Live Demo & Testing**
7. **Q&A and Next Steps**

What You Need

- Basic knowledge of the **C** programming language.
- Familiarity with the **command line** (Linux, macOS, or WSL).
- **Tools installed:** `flex` and `bison`.
 - Ubuntu/Debian: `sudo apt install flex bison`
 - macOS: `brew install flex bison`

Let's begin!